4.9 Discovering Relationships of Parallel and Perpendicular Lines

Lesson Introduction

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<tr><td>Benchmark</td><td colspan="2">MA.912.AR.2.3 Write a linear two-variable equation for a line that is parallel or perpendicular to a given line and goes through a given point.</td><td>Learning Targets</td><td colspan="2">I can discover the relationship between parallel lines.
I can discover the relationship between perpendicular lines.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="3">Working Toward</td></tr>
<tr><td colspan="2">MA.8.AR.4.3 Given a real-world context, solve systems of two linear equations by graphing.</td><td colspan="3">MA.912.GR.3.2 Given a mathematical or real-world context, use coordinate geometry to classify or justify definitions, properties and theorems involving circles, triangles, or quadrilaterals.</td></tr>
<tr><td>Essential Questions</td><td colspan="5">What does it mean when two lines have the same slope but different y-intercepts?
How are parallel lines different than perpendicular lines?
What key features determine if a line will be parallel or perpendicular to another line?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="5">Exploration drives this lesson on parallel and perpendicular lines. Students leverage their observations and questions with sets of parallel lines and sets of perpendicular lines to conceptualize key features of those lines, including their relationship with slope.</td></tr>
<tr><td rowspan="2">Component Summary</td><td>4.9.1 Warm-Up (~5 min)</td><td>Notice and Wonder routine on weight and balance.
(SE p. 208)</td><td>4.9.3 Exploration (15-25 min)</td><td colspan="2">Students will explore the concept of perpendicular lines and their key features.
(SE p. 211)</td></tr>
<tr><td>4.9.2 Exploration (15-25 min)</td><td>Students will explore the concept of parallel lines and their key features. (SE p. 209)</td><td>4.9.4 Wrap-Up (~5 min)</td><td colspan="2">Students complete a matching activity that assesses their ability to match two lines that are parallel and two lines that are perpendicular.
(SE p. 213)</td></tr>
<tr><td rowspan="2">Resources</td><td colspan="2">Glossary</td><td colspan="2">Materials</td><td>Additional Preparation</td></tr>
<tr><td colspan="2"><u>Key Terms:</u> N/A

<u>Additional Glossary:</u>
Parallel
Perpendicular
Reciprocal</td><td colspan="2">Straightedge
(Optional) Graphing Utility Activity
(Optional) Sticky Notes</td><td>N/A</td></tr>
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<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may mix up the relationship of slope for parallel and perpendicular lines.
Students may have trouble getting started when an equation is not in slope-intercept form, or interpret a key feature like y-intercept not in slope-intercept form.</td><td>Students are not graphing in this lesson. Instead, students use the graphs of lines to make observations and inferences about the properties of parallel and perpendicular lines.
Students will need fluency in converting between slope-intercept and standard form to access the connection between slope and features of lines.
The lesson contains two Explorations that are scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. Try to refrain from teaching the concepts and allow students to discover the relationships. This will help student understanding when you do teach these ideas in Lessons 10 and 11.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
The Warm-Up activity intentionally sets the stage for the observational approach to Lesson 9. 4.9.2 and 4.9.3 Explorations position students to make observations and ask questions that lead to key features of parallel and perpendicular lines.</td><td>MA.K12.MTR.5.1: Patterns and Structure
Exploration and inquiry-based learning inherently make use of patterns and connect them to mathematical concepts. For example, students may observe that parallel lines seem to never intersect and ultimately connect this the property of lines having the same slope. This standard focuses on students' ability to "…construct relationships between their current understanding and more sophisticated [mathematical] ways of thinking."</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">The framing of Notice and Wonder and 4.9.2 and 4.9.3 Explorations position students to explore a variety of low-floor, high-ceiling entry-points to conversations on parallel and perpendicular lines. Students have multiple entry-points.

Using an online graphing tool like Desmos and replacing the y-intercept while keeping the slope the same will stimulate connections and provide a visual entry-point. Also, consider how simulating the properties of Sets A, B, and C in 4.9.2 Exploration might open visual entry-points.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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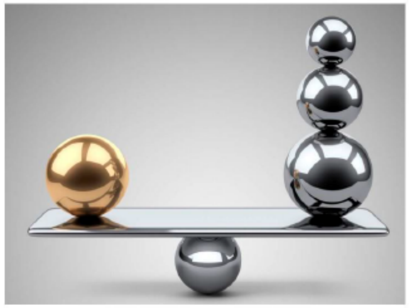
4.9.1 Warm-Up: Notice and Wonder

Component Overview: *Students complete a Notice and Wonder activity related to an image of weight and balance.*

4.9.1 Warm-Up: Notice and Wonder



1. What do you notice?
   Answers will vary. I notice 1 gold ball and 3 silver balls. I notice that the 1 gold ball weighs the same as the three silver balls.

2. What do you wonder?
   Answers will vary. The gold ball and silver ball look similar in size, so I wonder how they differ in weight. I wonder how they are all balancing without rolling off.

3. Write a question about the picture for the class.
   Answers will vary. Does the gold ball and the larger silver ball have any similar weights? How much do the smaller silver balls weigh?

Notice and Wonder
No wrong answers here – routines like Notice and Wonder intentionally foster equity in entry points for mathematical concepts using scaffolded prompts. 4.9.1 Warm-Up purposefully begins positioning students to be confident in sharing observations that will serve them in 4.9.2 and 4.9.3 Explorations of parallel and perpendicular lines.

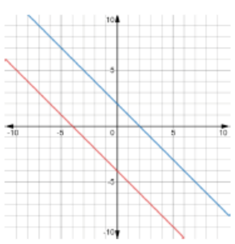
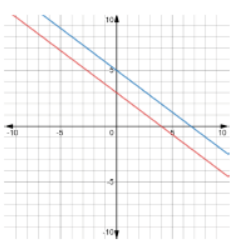
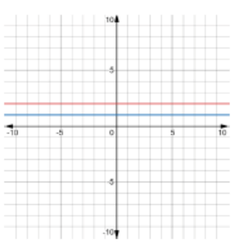
Consider having students share their questions using sticky notes. The teacher could choose one or two of the questions to share with the class.

4.9.2 Exploration: Parallel Lines

Component Overview: *Students will explore the concept of parallel lines and their key features.*

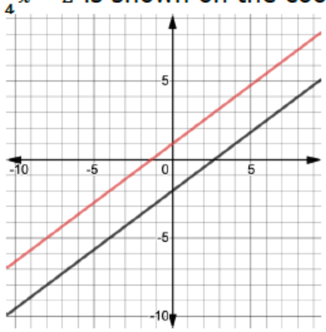
4.9.2 Exploration: Parallel Lines

1. Examine the pairs of parallel lines.
   Write down your observations and questions.

Parallel Lines Set A	Parallel Lines Set B	Parallel Lines Set C
		

Observations	Questions
Answers may vary. Lines in sets A, B, & C do not intersect. The lines appear to have the same slope.	*Answers may vary. Why don't the lines intersect? What does it mean when they don't intersect?*

2. The graph of $y = \frac{3}{4}x - 2$ is shown on the coordinate grid.



Do not lead students or front-load information in 4.9.2 Exploration. Instead, prompt students to make observations and ask questions as part of an inquiry process into learning, like a science experiment.

Consider utilizing the Desmos activity. Students can use the workbook to record their answers from Desmos.

If using the workbook, consider beginning by reviewing the context of the problem. Allow students to work with a partner to complete questions 1a, 1b, 1c together and then bring the whole class together to share their answers. Repeat this process for questions 2-4.

How is Set C the same as Sets A&B? How is it different? Do you think the quantitative relationship of the slopes in Set A and Set B would be different than the relationship between the slopes in Set C? Why or why not?

4.9.2 Exploration: Parallel Lines (continued)

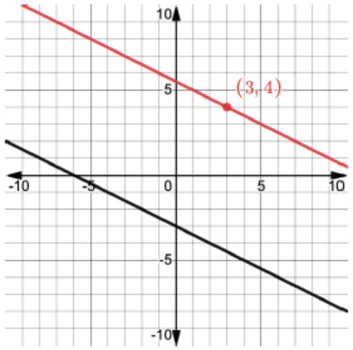
a. Draw a line that is parallel to $y = \frac{3}{4}x - 2$.
Answer should be the graph of a line with the same slope and different y-intercept. Sample answer shown.

b. Use the slope and the y-intercept of the line you drew to write the equation of a line that is parallel to $y = \frac{3}{4}x - 2$.
Answers will vary. Answer should have same slope and different y-intercept and should match what was graphed. Example from graph: $y = \frac{3}{4}x + 1$

c. Write the equations of the lines that two other classmates wrote.
Answers will vary. Equations should all have the same slope and different y-intercepts.

Classmate	Equation
Student name	$y = \frac{3}{4}x - \frac{1}{2}$
Student name	$y = \frac{3}{4}x + 12$

3. The graph of $y = -\frac{1}{2}x - 3$ is shown on the coordinate grid.



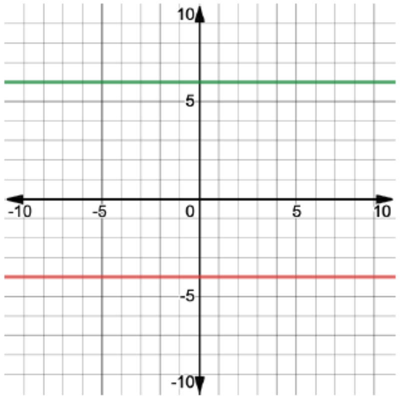
a. Sketch a line that is parallel to $y = -\frac{1}{2}x - 3$ and passes through the point $(3, 4)$.
Answer shown on graph.

Students should be familiar with parallel lines from earlier grade levels. They may not be familiar with graphing a parallel line. Allow for graphing of lines that “look” parallel on a graph. The expectation at this stage is not for students to know or use the exact definition of parallel lines. Students should be able to graph the equation provided and their parallel line should be one that doesn’t appear to intersect.

b. Write the equation of the line in point-slope form.

$(y-4)=-\frac{1}{2}(x-3)$

4. The graph of $y=6$ is shown on the coordinate grid.



a. Sketch a line that is parallel to $y=6$ and passes through the point $(-2,-4)$.

Answer graphed above.

b. Write the equation of the line.

$y=-4$

5. Write a conjecture about the slopes of parallel lines.

Answers will vary. Parallel lines have the same slope.

While circulating during 4.9.2 Exploration, listen for student talk. Consider using talk moves to facilitate discussion and extend student thinking to produce increasingly precise mathematical reasoning.

- *“They are the same.” (What are the same? Why are they the same? Why is that an important/necessary feature for parallel lines?)*
- *“Different intercepts.” (Will parallel lines always have different intercepts? Why or why not? Give an example.)*

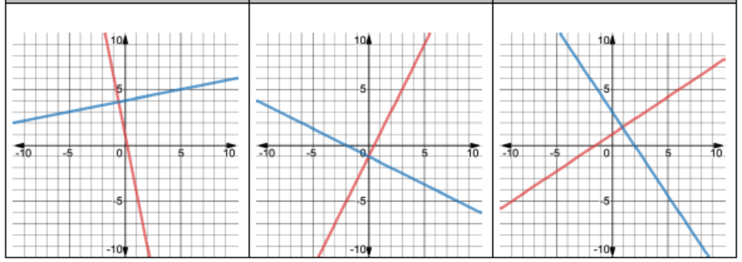
4.9.3 Exploration: Perpendicular Lines

Component Overview: *Students will explore the concept of perpendicular lines and their key features.*

4.9.3 Exploration: Perpendicular Lines

1. Examine the pairs of perpendicular lines.

Perpendicular Lines Set A	Perpendicular Lines Set B	Perpendicular Lines Set C



a. Write down your observations and questions.

Observations	Questions
Answers will vary. Sets A,B,&C have two lines that intersect. There is one point that both lines share. There is one line with a positive slope and one with a negative slope in each set.	Answers will vary. Why do these lines intersect? What are the slopes of each line? Do they intersect at 90 degrees if they are perpendicular?

b. Record your partner's observations and questions.

Observations	Questions
Answers will vary.	Answers will vary.

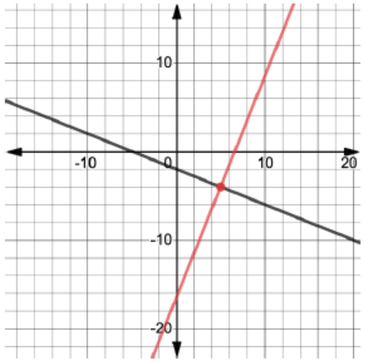
For this activity, begin by reviewing the context of the problem. Allow students to work with a partner to complete questions 1a, 1b, and 1c together and then bring the whole class together to share their answers. Repeat this process for questions 2-4.

Are students struggling with finding how best to make observations and create question prompts? Consider using sentence stems to target certain concepts:

- *I noticed that both lines _________ which makes me wonder if ________.*
- *I noticed the red line is always ______ and it makes me wonder if the slope is ______ compared to the slope of the blue line.*
- *I notice the intersection of the two lines makes the shape of _______ and it makes me wonder if their slopes are _________.*

Students are not just copying what their partner writes but instead critically listening and taking notes. Students record a partner's observation for an additional entry-point and scaffold when making sense of observations and questions.

2. The graph of $y = -\frac{2}{5}x - 2$ is shown on the coordinate grid.

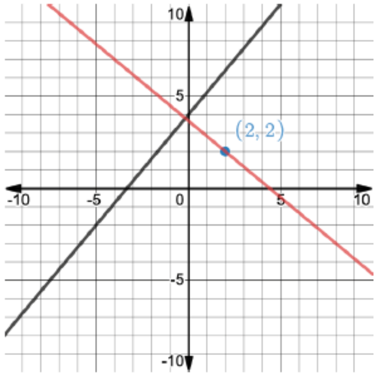


   a. Draw a line that is perpendicular to $y = -\frac{2}{5}x - 2$ and goes through the point $(5, -4)$ by using the corner of a sheet of paper for a right angle. *Answer on graph.*

   b. Use your observations and the line you drew to determine the slope that completes the equation of the line that is perpendicular to $y = -\frac{2}{5}x - 2$ and passes through the point $(5, -4)$.

$$y + 4 = \frac{5}{2}(x - 5)$$

3. The graph of $6x - 5y = -20$ is shown on the coordinate grid.



   a. Draw a line that is perpendicular to $6x - 5y = -20$ and goes through the point $(2, 2)$ by using the corner of a sheet of paper for a right angle. *Answer on graph.*

   b. Use your observations and the line you drew to determine the slope that completes the equation of the line that is perpendicular to $6x - 5y = -20$ and passes through the point $(2, 2)$.

$$y - 2 = -\frac{5}{6}(x - 2)$$

Kinesthetic learners will benefit from question 2a. Show students how to use the corner of a sheet of paper for a right angle. Consider leveraging technology, such as a document camera, to show students the process for doing this with a sheet of paper so that students can better understand how to do this step and assist visual learners conceptually with what is occurring.

Students do not yet formalize their conjectures about the slope of perpendicular lines.

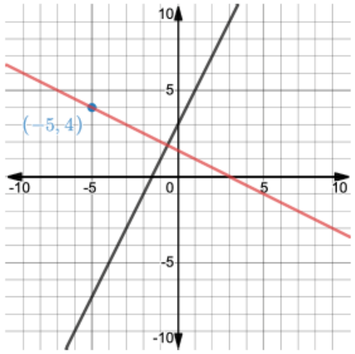
While circulating during the exploration, listen for student talk. If necessary, position students to make connections between each slope.

- *What do you notice about the slope of the new line?*
- *How is it the same or different than the slope of the first line?*
- *Do we think this is always true? Let's explore more.*

Consider an additional entry-point here for students who struggle with standard form. The spiraling for fluency of converting between forms is important and embedded within this lesson, but include optional entry points for slope-intercept form to remove any barriers from the learning target of this lesson.

4.9.3 Exploration: Perpendicular Lines (continued)

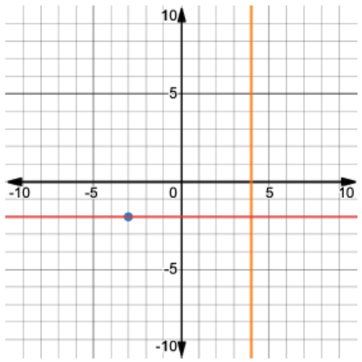
4. The graph of $y - 5 = 2(x - 1)$ is shown on the coordinate grid.



   a. Draw a line that is perpendicular to $y - 5 = 2(x - 1)$ and goes through the point $(-5, 4)$ by using the corner of a sheet of paper for a right angle. *Answer on graph.*

   b. Write the equation of the line that is perpendicular to $y - 5 = 2(x - 1)$ at the point $(-5, 4)$.
   $y - 4 = -\frac{1}{2}(x + 5)$

5. The graph of $x = 4$ is shown on the coordinate grid.



   a. Draw a line that is perpendicular to $x = 4$ and goes through the point $(-3, -2)$. *Answer on the graph.*

   b. Write the equation of the line that is perpendicular to $x = 4$ that passes through the point $(-3, -2)$.
   $y = -2$

6. Write a conjecture about the slopes of perpendicular lines.
   Answers will vary. Perpendicular lines have slopes that are opposites and reciprocals.

MA.K12.MTR.5.1: Patterns and Structure
Continue probing students to make use of the pattern and structure of the slope for each equation. Probe students to make connections between the concrete calculations of slope and the abstract relationship between perpendicular lines.

If time permits, have students revisit this set of lines after formalizing the relationship between the slope of perpendicular lines. The slopes of zero and undefined will challenge students to think critically about the relationship in a more complex context.

While circulating during the exploration, listen for student talk. Elicit student responses with talk moves that challenge their thinking and work toward increasingly precise responses.

- *"They are the same but opposite." (What does your fellow student mean by "the same"? What does she mean by "the opposite"?)*
- *"You can just flip the slope and make it the opposite sign." (What do you mean by "flip"? What if the equation is in standard form? How would I know?)*

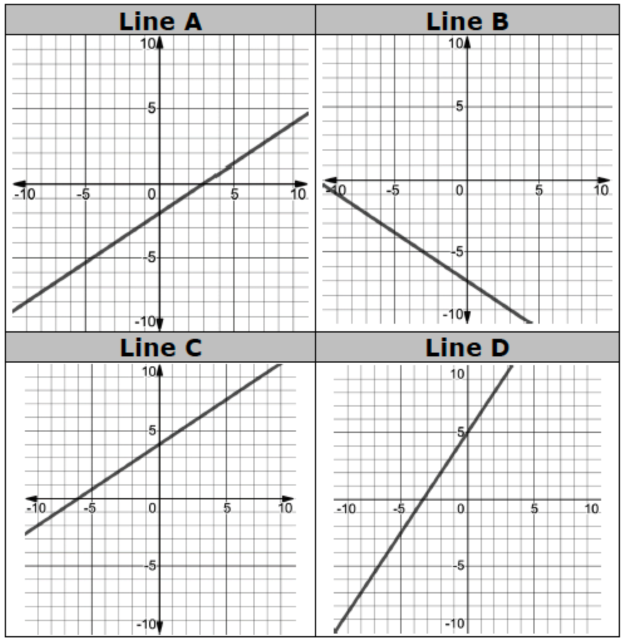
4.9.4 Wrap-Up: Ticket Out the Door

Component Overview: *Students will complete a matching activity that assesses their ability to match two lines that are parallel and two lines that are perpendicular.*

4.9.4 Wrap-Up: Ticket Out the Door

1. Match the two lines that are parallel and the two lines that are perpendicular. Then, match the equations.

	Lines	Equations
Parallel	*A & C*	*E & G*
Perpendicular	*B & D*	*F & H*



Equation E	Equation F
$y + 2 = \frac{2}{3}(x + 9)$	$2x + 3y = -21$
Equation G	**Equation H**
$-2x + 3y = -6$	$y = \frac{3}{2}x + 5$

Proficiency on 4.9.4 Wrap-Up includes:

- *Comparing parallel and perpendicular lines*
- *Fluency of use of standard form, slope-intercept form and point-slope form and the conversion between these forms*

Lesson 10 covers writing the equations of parallel lines. Consider adding a challenge question for students to write the equation of a line parallel to Line A for a glimpse into your students' entry-points to the next lesson.